

OpenStax Calculus Volume 3

Section 4.1 Exercises: Functions of Several Variables

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For the following exercises, evaluate each function at the indicated values.

1. $W(x, y) = 4x^2 + y^2$. Find $W(2, -1)$, $W(-3, 6)$.
2. $W(x, y) = 4x^2 + y^2$. Find $W(2 + h, 3 + h)$.
3. The volume of a right circular cylinder is calculated by a function of two variables, $V(x, y) = \pi x^2 y$, where x is the radius of the right circular cylinder and y represents the height of the cylinder. Evaluate $V(2, 5)$ and explain what this means.
4. An oxygen tank is constructed of a right cylinder of height y and radius x with two hemispheres of radius x mounted on the top and bottom of the cylinder. Express the volume of the tank as a function of two variables, x and y , find $V(10, 2)$, and explain what this means.

For the following exercises, find the domain of the function.

5. $V(x, y) = 4x^2 + y^2$
6. $f(x, y) = \sqrt{x^2 + y^2 - 4}$
7. $f(x, y) = 4 \ln(y^2 - x)$
8. $g(x, y) = \sqrt{16 - 4x^2 - y^2}$
9. $z(x, y) = y^2 - x^2$
10. $f(x, y) = \frac{y+2}{x^2}$

Find the range of the functions.

11. $g(x, y) = \sqrt{16 - 4x^2 - y^2}$
12. $V(x, y) = 4x^2 + y^2$
13. $z = y^2 - x^2$

For the following exercises, find the level curves of each function at the indicated value of c to visualize the given function.

14. $z(x, y) = y^2 - x^2$, $c = 1$

15. $z(x, y) = y^2 - x^2, c = 4$
16. $g(x, y) = x^2 + y^2, c = 4, c = 9$
17. $g(x, y) = 4 - x - y, c = 0, 4$
18. $f(x, y) = xy, c = 1, c = -1$
19. $h(x, y) = 2x - y, c = 0, -2, 2$
20. $f(x, y) = x^2 - y, c = 1, 2$
21. $g(x, y) = \frac{x}{x+y}, c = -1, 0, 2$
22. $g(x, y) = x^3 - y, c = -1, 0, 2$
23. $g(x, y) = e^{xy}, c = \frac{1}{2}, 3$
24. $f(x, y) = x^2, c = 4, 9$
25. $f(x, y) = xy - x, c = -2, 0, 2$
26. $h(x, y) = \ln(x^2 + y^2), c = -1, 0, 1$
27. $g(x, y) = \ln(\frac{y}{x^2}), c = -2, 0, 2$
28. $z = f(x, y) = \sqrt{x^2 + y^2}, c = 3$
29. $f(x, y) = \frac{y+2}{x^2}, c = \text{any constant}$

For the following exercises, find the vertical traces of the functions at the indicated values of x and y , and plot the traces.

30. $z = 4 - x - y, x = 2$
31. $f(x, y) = 3x + y^3, x = 1$
32. $z = \cos \sqrt{x^2 + y^2}, x = 1$

Find the domain of the following functions.

33. $z = \sqrt{100 - 4x^2 - 25y^2}$
34. $z = \ln(x - y^2)$
35. $f(x, y, z) = \frac{1}{\sqrt{36 - 4x^2 - 9y^2 - z^2}}$
36. $f(x, y, z) = \sqrt{49 - x^2 - y^2 - z^2}$
37. $f(x, y, z) = \sqrt[3]{16 - x^2 - y^2 - z^2}$
38. $f(x, y) = \cos \sqrt{x^2 + y^2}$

For the following exercises, plot a graph of the function.

39. $z = f(x, y) = \sqrt{x^2 + y^2}$
40. $z = x^2 + y^2$
41. Use technology to graph $z = x^2y$.

Sketch the following by finding the level curves. Verify the graph using technology.

42. $f(x, y) = \sqrt{4 - x^2 - y^2}$

43. $f(x, y) = 2 - \sqrt{x^2 + y^2}$

44. $z = 1 + e^{-x^2 - y^2}$

45. $z = \cos \sqrt{x^2 + y^2}$

46. $z = y^2 - x^2$

47. Describe the contour lines for several values of c for $z = x^2 + y^2 - 2x - 2y$.

Find the level surface for the functions of three variables and describe it.

48. $w(x, y, z) = x - 2y + z, c = 4$

49. $w(x, y, z) = x^2 + y^2 + z^2, c = 9$

50. $w(x, y, z) = x^2 + y^2 - z^2, c = -4$

51. $w(x, y, z) = x^2 + y^2 - z^2, c = 4$

52. $w(x, y, z) = 9x^2 - 4y^2 + 36z^2, c = 0$

For the following exercises, find an equation of the level curve of f that contains the point P .

53. $f(x, y) = 1 - 4x^2 - y^2, P(0, 1)$

54. $g(x, y) = y^2 \arctan x, P(1, 2)$

55. $g(x, y) = e^{xy}(x^2 + y^2), P(1, 0)$

56. The strength E of an electric field at point (x, y, z) resulting from an infinitely long charged wire lying along the z -axis is given by $E(x, y, z) = k/\sqrt{x^2 + y^2}$, where k is a positive constant. For simplicity, let $k = 1$ and find the equations of the level surfaces for $E = 10$ and $E = 100$.

57. A thin plate made of iron is located in the xy -plane. The temperature T in degrees Celsius at a point $P(x, y)$ is inversely proportional to the square of its distance from the origin. Express T as a function of x and y .

58. Refer to the preceding problem. Using the temperature function found there, determine the proportionality constant if the temperature at point $P(1, 2)$ is 50°C . Use this constant to determine the temperature at point $Q(3, 4)$.

59. Refer to the preceding problem. Find the level curves for $T = 40^\circ\text{C}$ and $T = 100^\circ\text{C}$, and describe what the level curves represent.